AmberMD Chatbot

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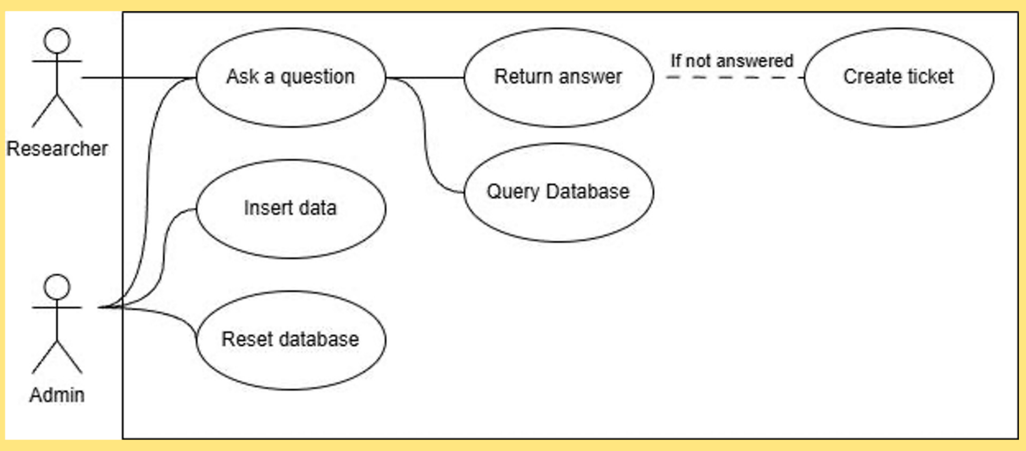
Background

Amber Molecular Dynamics (AmberMD) offers a suite of biomolecular simulation programs, and a discussion forum (Amber Mail Reflectors) for users to discuss and present any questions regarding the software and any bug fixes. However, not every question is answered in the Amber Reflector, instead spread out between additional forums, tutorials, and mailing archives.

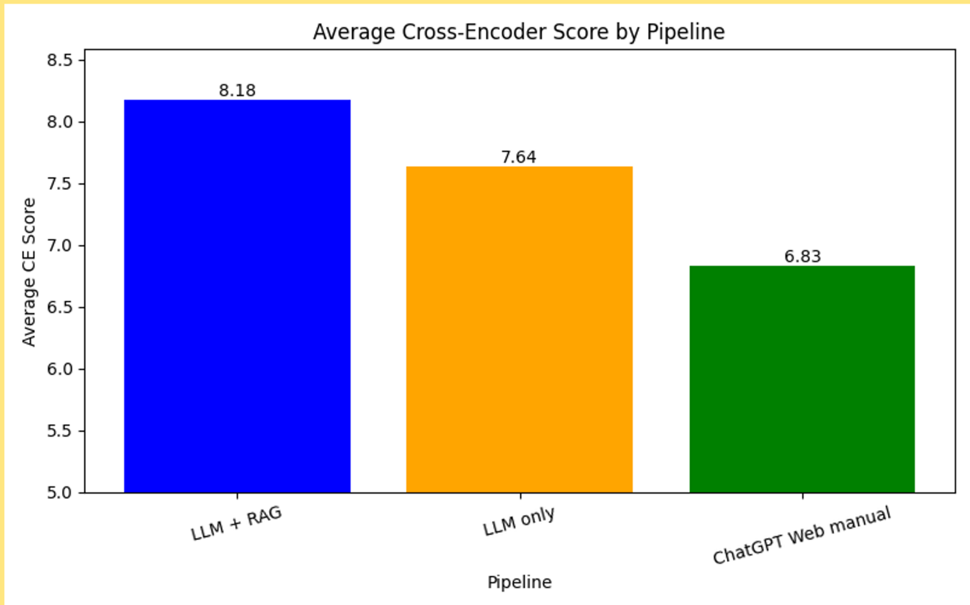
Objective

Our goal is to create a chatbot experience trained on the various collections of AmberMD resources; to provide clear and concise answers, complete with links to the original sources and distinctions between official and AI supported answers, to users of the Amber software.

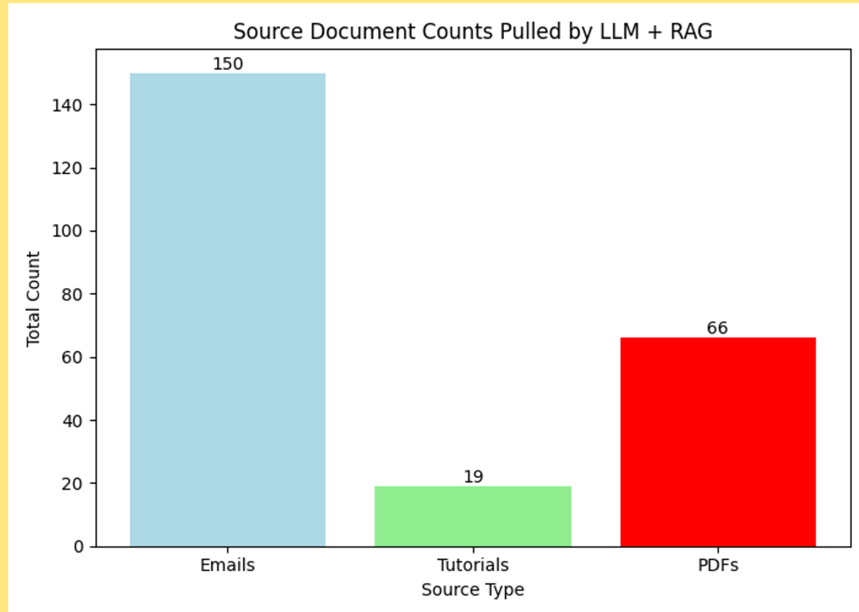


Tools and Technology

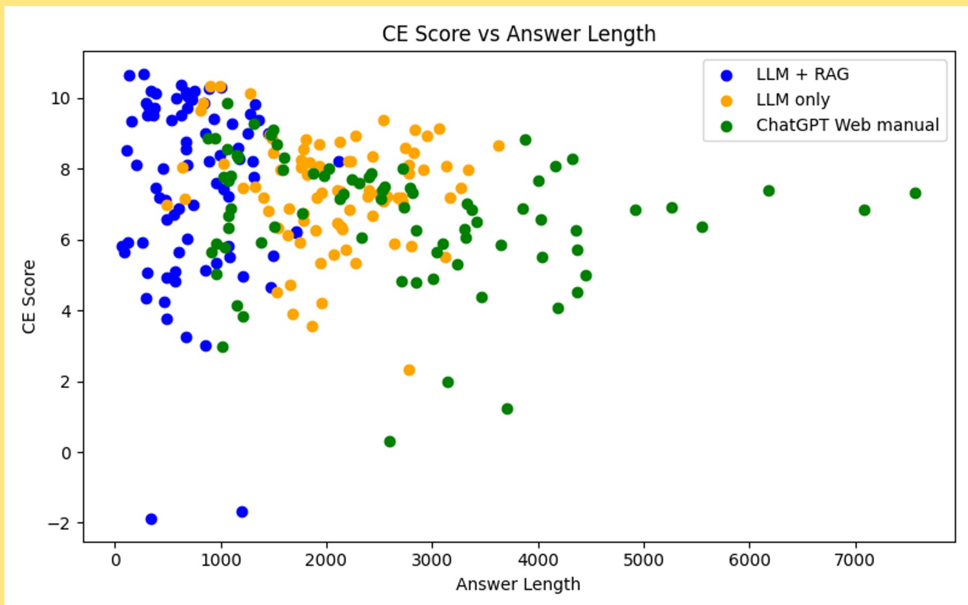
Apache Airflow, socket.io, python, Hugging Face, BeautifulSoup, chroma, jupyter



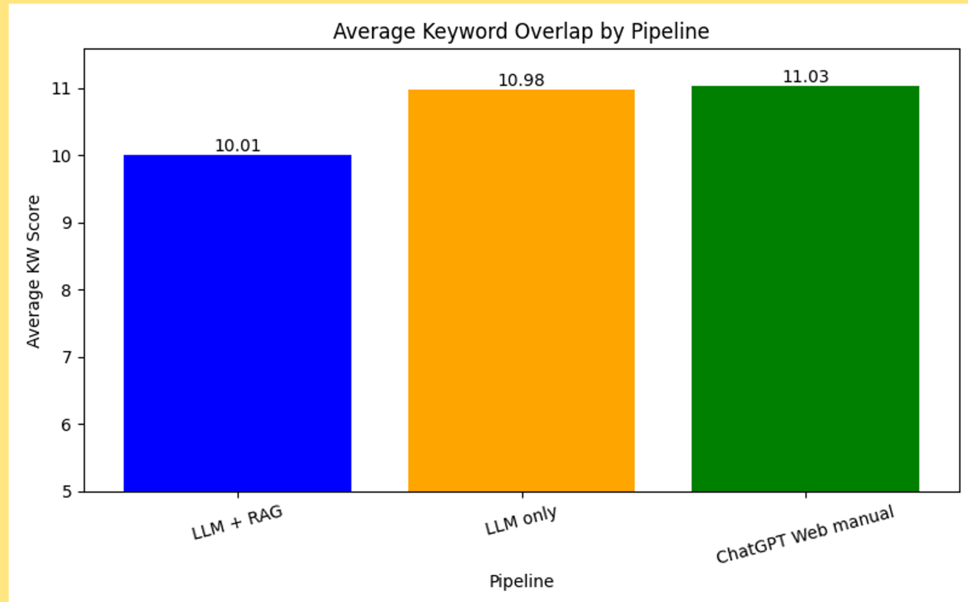
[Example 1- Average Cross-Encoder Score]



[Example 4- Source Document Pull Count]



[Example 2- CE Score vs Answer Length]



[Example 3- Average Keyword Overlap]

Example Explanations

Example 1

- Cross Encoder similarity test concludes LLM + RAG has the highest accurate results. Provided context the LLM + RAG is promising in achieving it's goal.

Example 2

- CE Score vs Answer Length test LLM + RAG has highest similarity results but less length for final output. Results in accurate answers but with less amount of words.

Example 3

- Average Keyword Overlap test LLM + RAG has the lowest keyword overlap score. ChatGPT provides more keyword overlap without provided context than LLM + RAG.

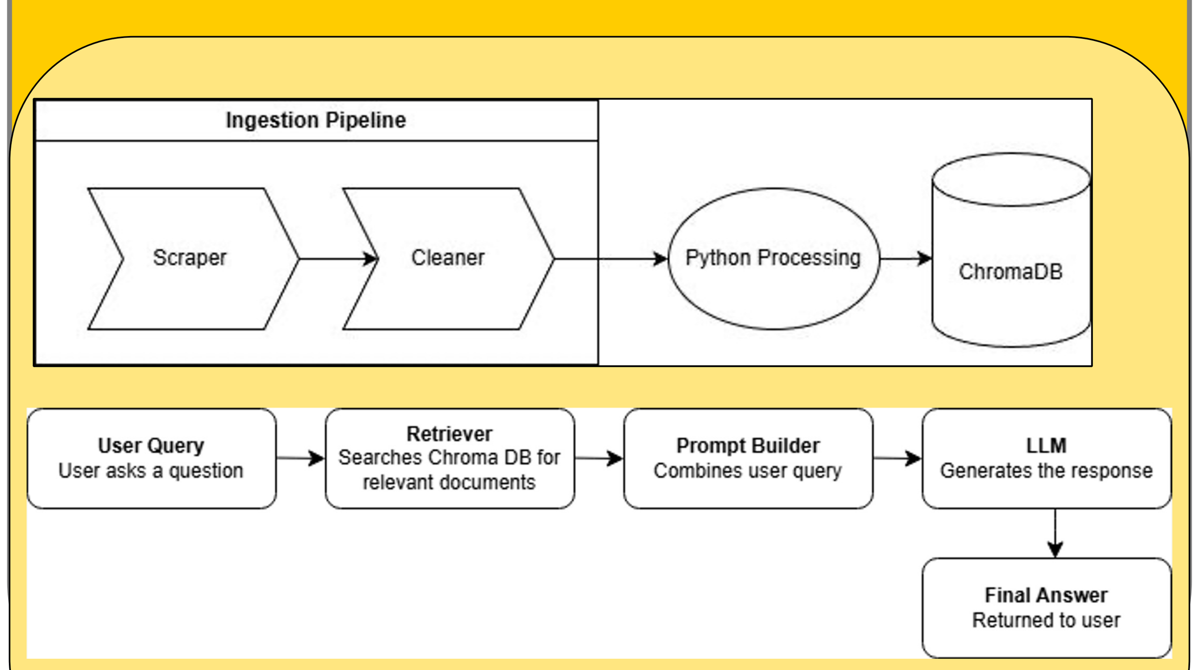
Example 4

- Sources Pulled by LLM + RAG results in more sources used by Emails rather than Tutorials and PDFs. Email's dating back from 2020 have more information to work with than tutorials and PDFs.

Results

Our data shows that the LLM + RAG pipeline performs better than other pipelines in result accuracy and similarity while having less length to its answers. This would suggest that the pipeline is optimized for shorter, direct answers, at the cost of a lower keyword overlap compared to the ChatGPT pipeline. The LLM + RAG pipeline is also consistent in using email data to formulate answers, while count of PDFs and tutorials pulled vary by inquiry.

High level overview



Conclusion

Our team accomplished:

- Implementing an automatic retrieval of AmberMD data through scraping its guides, forums, and mailing archives
- Designing a database capable of securely storing AmberMD data
- Training a chatbot through analyzing data vector similarity comparisons
- Assembling Amber tutorials through use of container environment images